EXPERIMENT: GYM FITNESS TRACKER VISUALIZATION

AIM:

To create data visualizations such as pie charts and bar graphs to analyze workout activities and calories burned using JavaScript and Chart.js.

OBJECTIVE:

• To understand data visualization techniques
• To represent fitness data using charts
• To use Chart.js for creating interactive graphs
• To display values inside a pie chart

INTRODUCTION:

Data Visualization is the graphical representation of data using charts and graphs. In this experiment, a Gym Fitness Tracker is developed to visualize workout data such as calories burned across different activities like Cardio, Strength Training, Yoga, HIIT, and Cycling.

PROCEDURE:

Step 1: Create HTML File (index.html)
• Create a basic HTML structure
• Add canvas elements for charts
• Include Chart.js and datalabels plugin

Step 2: Create JavaScript File (script.js)
• Define workout data
• Configure pie chart with values inside
• Configure bar chart

Step 3: Run the Project
• Open index.html in a browser

CODE:

Refer to index.html and script.js files created using Chart.js and datalabels plugin.

```html
<!DOCTYPE html>

<html lang="en">

<head>

    <meta charset="UTF-8">

    <meta name="viewport" content="width=device-width,
initial-scale=1.0">

    <title>Gym Fitness Tracker</title>


    <style>

        body {

            font-family: Arial;

            text-align: center;

            margin: 40px;

            background-color: #f4f6f9;

        }


        h1 {

            color: #333;

        }


        canvas {

            margin: 20px auto;

            display: block;

            background: white;

            padding: 15px;
```

```
            border-radius: 12px;

            box-shadow: 0 0 10px rgba(0,0,0,0.1);

        }

    </style>

</head>

<body>



    <h1>Gym Fitness Tracker</h1>



    <!-- Pie Chart -->

    <canvas id="pieChart" width="400" height="400"></canvas>



    <!-- Bar Chart -->

    <canvas id="barChart" width="400" height="400"></canvas>



    <!-- Chart.js -->

    <script src="https://cdn.jsdelivr.net/npm/chart.js"></script>



    <!-- Plugin for showing values -->

    <script
src="https://cdn.jsdelivr.net/npm/chartjs-plugin-datalabels"></scrip
t>



    <!-- JS File -->

    <script src="script.js"></script>
```

```
</body>

</html>
```

```
// Register plugin

Chart.register(ChartDataLabels);


// Fitness Data (Calories Burned)

const fitnessData = {

    labels: ['Cardio', 'Strength', 'Yoga', 'HIIT', 'Cycling'],

    datasets: [{

        label: 'Calories Burned',

        data: [300, 450, 200, 350, 250],

        backgroundColor: [

            '#ff6384',

            '#36a2eb',

            '#ffce56',

            '#4bc0c0',

            '#9966ff'

        ]

    }]

};


// PIE CHART (with values inside)

const pieCtx = document.getElementById('pieChart').getContext('2d');
```

```
new Chart(pieCtx, {
    type: 'pie',
    data: fitnessData,
    options: {
        plugins: {
            title: {
                display: true,
                text: 'Workout Distribution (Calories)'
            },
            datalabels: {
                color: '#fff',
                font: {
                    weight: 'bold',
                    size: 14
                },
                formatter: (value) => value + ' cal'
            }
        }
    }
});

// BAR CHART
const barCtx = document.getElementById('barChart').getContext('2d');
```

```
new Chart(barCtx, {
    type: 'bar',
    data: fitnessData,
    options: {
        plugins: {
            title: {
                display: true,
                text: 'Calories Burned by Workout'
            }
        },
        scales: {
            y: {
                beginAtZero: true
            }
        }
    }
});
```

OUTPUT:

- Pie chart showing calorie distribution (values inside chart)

• Bar chart comparing workout activities

RESULT:

Thus, the Gym Fitness Tracker visualization was successfully created using JavaScript and Chart.js with values displayed inside the pie chart.